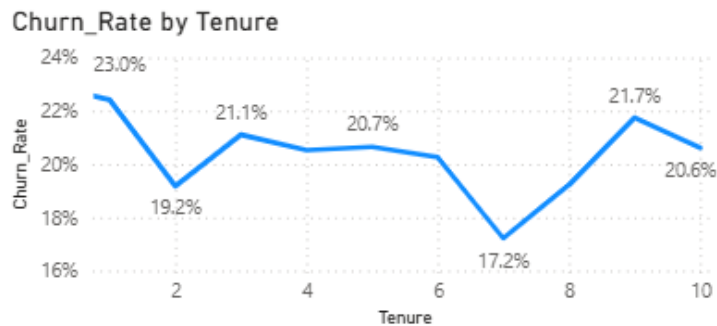
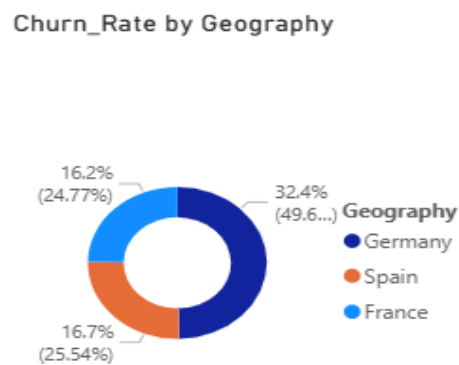


Section A: Power BI – Business Intelligence & Dashboards

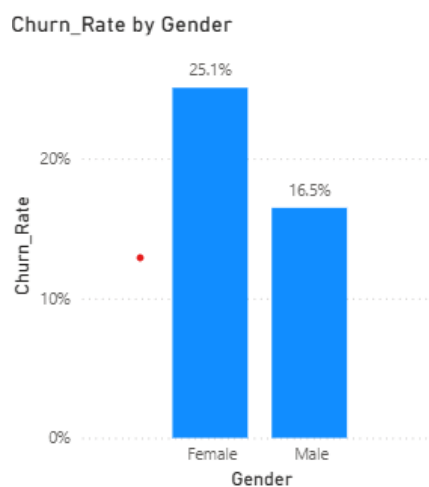
1. What is the overall churn rate, and how does it trend over time (if applicable)?



2. How does churn vary by geography (France, Germany, Spain)?

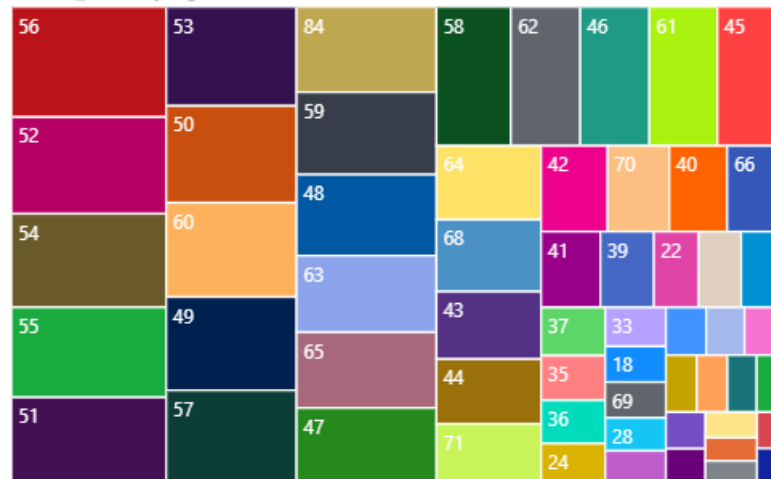


3. What are the differences in churn rates between male and female customers?



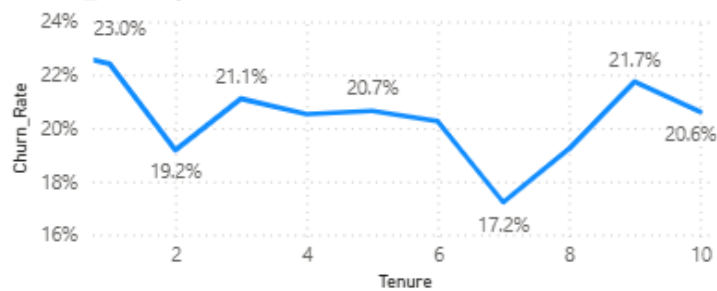
4. What is the churn rate distribution across different age groups?

Churn_Rate by Age



5. How does churn behavior change with tenure?

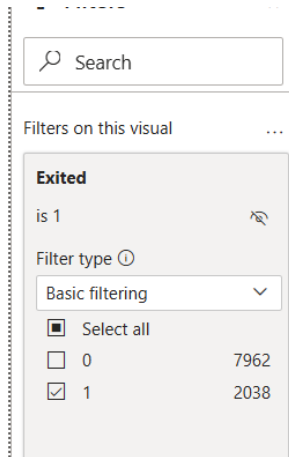
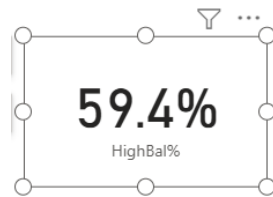
Churn_Rate by Tenure



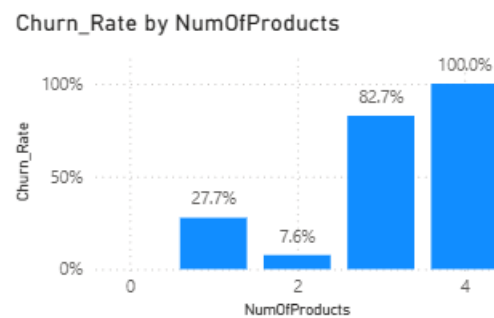
6. What percentage of high-balance customers (e.g., >\$100K) have churned?

```
1 High_balance = IF('Customer-Churn-Records'[Balance]>=100000,"High Balance","Low Balance")
```

```
1 High_Bal_CustCount = CALCULATE(DISTINCTCOUNT('Customer-Churn-Records'[CustomerId]),'Customer-Churn-Records'[High_balance]="High Balance")
```

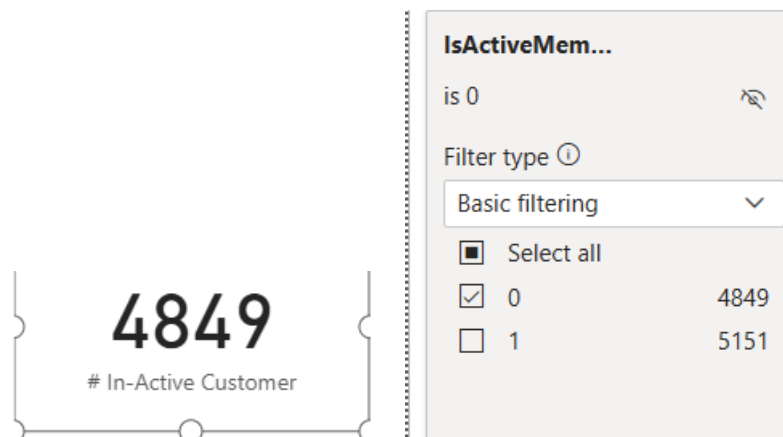


7. How do churn rates compare across different NumOfProducts?



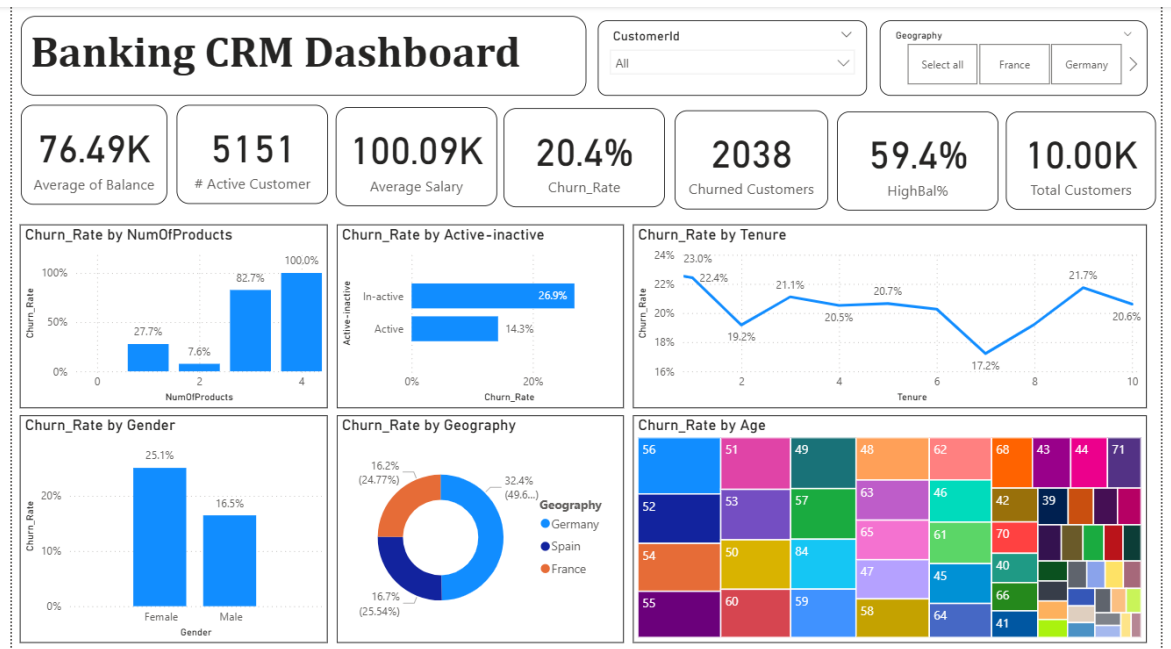
8. Are inactive members (IsActiveMember = 0) more likely to churn?

In-Active Customer = `if('Customer-Churn-Records'[IsActiveMember]=0,'Customer-Churn-Records'[CustomerId])`



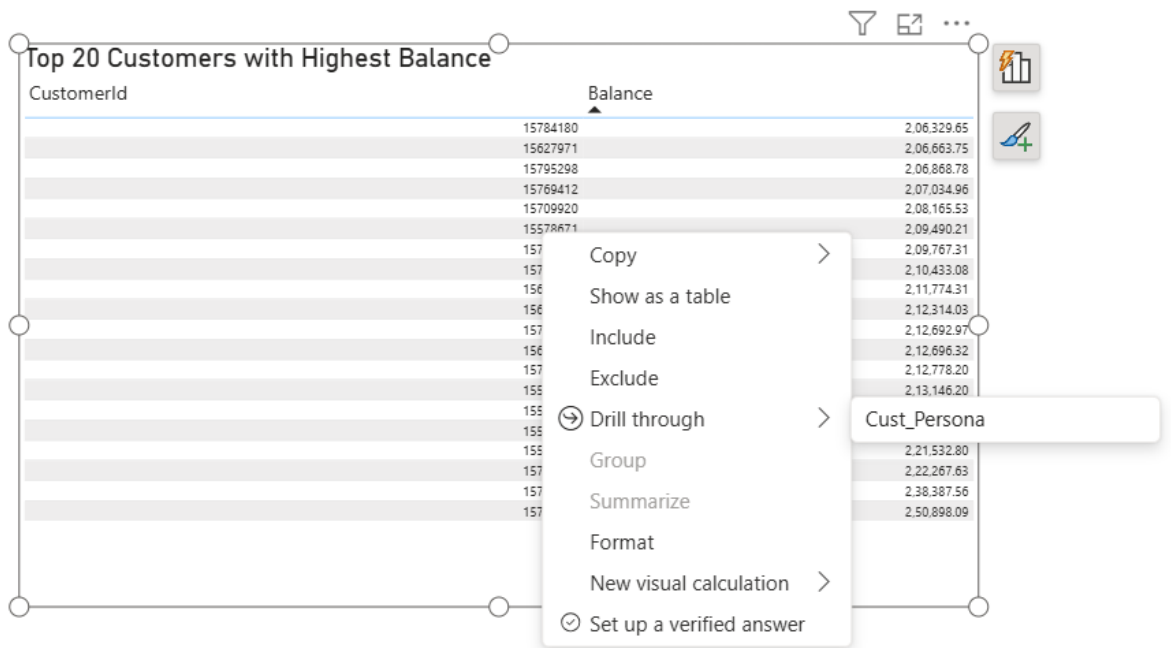
The inactive members are more likely to churn . 4849 are inactive here .

9. Can we build a KPI dashboard to track churn metrics like average salary, balance, and active users?



The above dashboard visual represents churn rate in various aspects (Churn Rate by no.of prod, churn rate by active-inactive cust, churn rate by tenure , churn rate by geography , churn rate by age , churn rate by gender . 2 slicers are present allowing selection of customer id and geography via which we can apply filters and check data accordingly. Also it showcases KPI card representing Average Balance, Active Customers , Average Salary , Churn Rate , Churned Customers , High Balance Customer Percent and Total Customers.

10. What does a customer persona dashboard (churned vs. retained) look like?



Customer_Id	Surname	Point Earned	Card Type	ChurnStatus	Age	Gender	Geography	Balance	EstimatedSalary	CreditScore
15578671	Webb	243	DIAMOND	Churned	29	Female	Spain	2,09,490.21	1,33,267.69	706

Based on the selection of Customer Id , Customer’s data is drilled through and its all information and churn status is shown in the below table on the Cust_Persona page.

Customer_Id	Surname	Point Earned	Card Type	ChurnStatus	Age	Gender	Geography	Balance	EstimatedSalary	CreditScore
15714241	Haddon	763	GOLD	Churned	42	Male	Spain	2,22,267.63	1,01,108.85	749

Filters

Search

Filters on this page

CustomerId

is 15714241

Top 20 CustomerId by Su...

Add data fields here

Filters on all pages

Add data fields here

Format

Search

Page information

Name

Cust_Persona

Page type

Drillthrough

Keep all filters

On

Cross-report

Off

Drill through from

CustomerId

+Add data

Drill through when

CustomerId is

Used as category

Allow Q&A

Off

Page synonyms

Add terms that might be used to refer to this page. Separate your list

Page 1

Page 2

Page 3

Details

Cust_Persona

Section B: Python – Exploratory Data Analysis (Provide a description of your answers in all the cells explaining the outputs)

11. Check the percentage of missing data and handle accordingly.

Now find the null values and calculate the total sum of it

```
[14]: df.isnull().sum()
```

```
[14]: RowNumber      0
      CustomerId    0
      Surname       0
      CreditScore   0
      Geography     0
      Gender        0
      Age           0
      Tenure        0
      Balance       0
      NumOfProducts 0
      HasCrCard     0
      IsActiveMember 0
      EstimatedSalary 0
      Exited        0
      Complain      0
      Satisfaction Score 0
      Card Type     0
      Point Earned  0
      dtype: int64
```

as found there are no missing values in the data .

12. How many rows and columns are there in the dataset?

```
[12]: #calculate the rows and column

      df.shape
```

```
[12]: (10000, 18)
```

There are 1000 rows and 18 columns in the dataset .

13. What is the distribution of churned vs. non-churned customers?

```
[17]: #Now showing the distribution of churned vs non-churned customers
```

```
churn_distr=df['Exited'].value_counts()
print('Churned Vs Non Churned Status')
print(churn_distr)
```

```
Churned Vs Non Churned Status
Exited
0      7962
1      2038
Name: count, dtype: int64
```

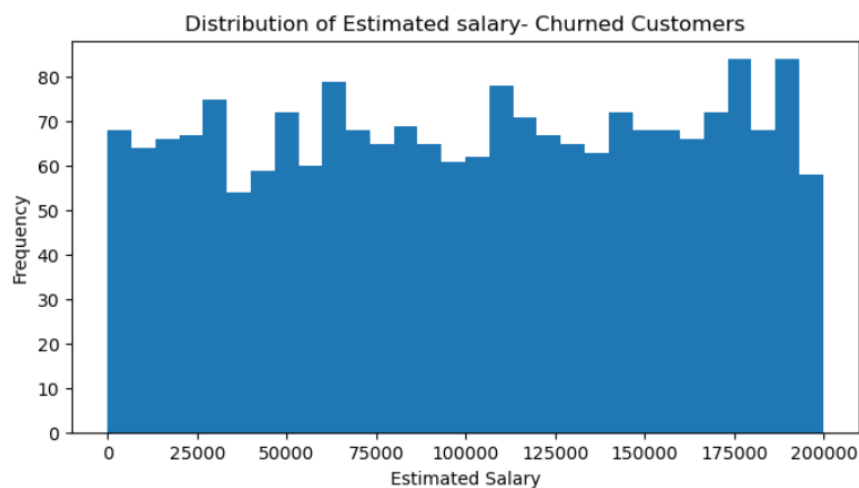
```
[ ]: 2038 customers have churned . 7962 are non-churned customers .
```

14. What is the distribution of EstimatedSalary of churned and retained Customers?

```
[115]: # to showcase the visualisation
import matplotlib.pyplot as plt
churned=df[df['Exited']==1]['EstimatedSalary']
retained=df[df['Exited']==0]['EstimatedSalary']

#Code for plotting the distribution of churned customers

plt.figure(figsize=(8,4))
plt.hist(churned,bins=30)
plt.xlabel("Estimated Salary")
plt.ylabel("Frequency")
plt.title("Distribution of Estimated salary- Churned Customers")
plt.show()
```



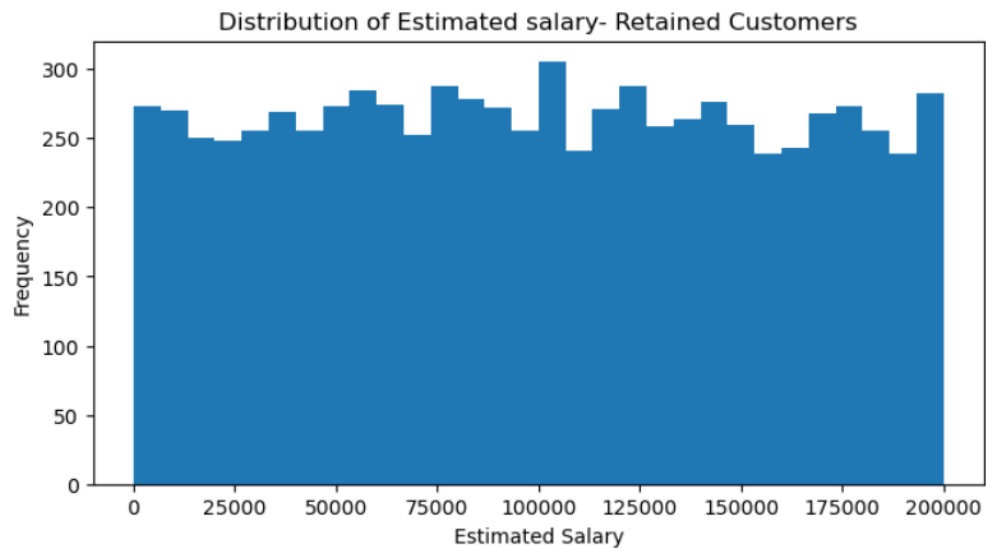
In the above output , distribution of churned customers is shown.

The churned customers here mostly lie between the salary range of 175000-200000. And the other range where most churned customers exist lie within the range of 60000-75000 & 100000-120000.


```
[114]: # Now to calculate the estimated Salary of retained customers(visualization)
```

```
#Code for plotting the distribution of retained customers
```

```
plt.figure(figsize=(8,4))
plt.hist(retained,bins=30)
plt.xlabel("Estimated Salary")
plt.ylabel("Frequency")
plt.title("Distribution of Estimated salary- Retained Customers")
plt.show()
```



```
[ ]: Here the most of the retained customers lie within the salary range of 100000-115000.
```

15. How do churn rates vary by Gender, Geography, and IsActiveMember?

```
[39]: #To calculate Churn Rates that vary by Gender,Geography,and IsActiveMember
#Calculating mean to understand the average of each .(Finding churn rates in percentage)
churn_by_gender=df.groupby('Gender')['Exited'].mean()*100
print(churn_by_gender)
```

```
Gender
Female    25.071539
Male      16.474253
Name: Exited, dtype: float64
```

```
[ ]: 25% churn rate exist for females and 16.4 % Churn rate exist for male customers.
```

```
[40]: churn_by_geography=df.groupby('Geography')['Exited'].mean()*100
print(churn_by_geography)
```

```
Geography
France    16.174711
Germany   32.443204
Spain     16.673395
Name: Exited, dtype: float64
```

```
16.1% churn rate in France , 32.44 % churn rate in Germany , 16.67 % churn rate in Spain .
```

```
[41]: churn_by_active=df.groupby('IsActiveMember')['Exited'].mean()*100
print(churn_by_active)
#Active->1 Not Active->0
```

```
IsActiveMember
0     26.871520
1     14.269074
Name: Exited, dtype: float64
```

```
14.2% active customers are churned . 26.8% inactive customers are not churned .
```

16. What is the average CreditScore, Balance, and EstimatedSalary of churned vs. retained customers?

```
•[47]: #Calculate Average CreditScore,Balance, and Estimated Salary for churned vs retained customers
#Using agg() calculating average of each .
avg_metrics=df.groupby('Exited').agg({'CreditScore':'mean','Balance':'mean','EstimatedSalary':'mean'})
print(avg_metrics)
```

	CreditScore	Balance	EstimatedSalary
Exited			
0	651.837855	72742.750663	99726.853141
1	645.414622	91109.476006	101509.908783

The above output displays the average of both churned and retained customers for Credit score ,Blanace and Estimated Salary where
Average of CreditScore for churned customers is 645.4 and same for retained customer is 651.8
Average of balance for churned Customers is 91109.4 and same for retained customer is 72742.75
Average of Estimated Salary for churned customers is 101509.908 and same for retained customer is 99726.853

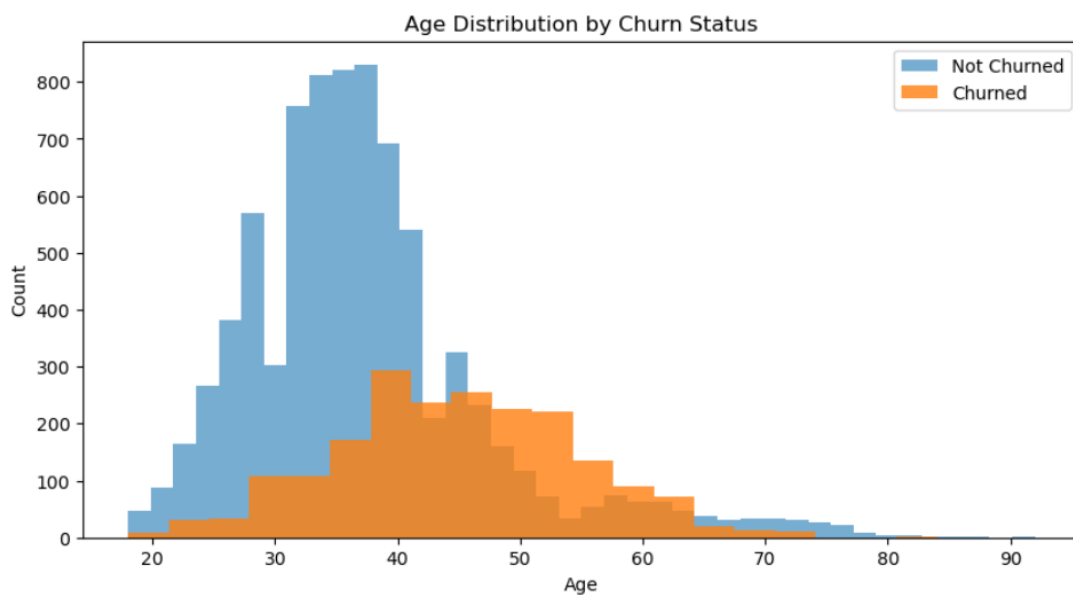
17. How does Age impact churn? Plot histograms and boxplots for churned and non-churned groups.

```
[124]: # Impact of Age on Churn Data

import matplotlib.pyplot as plt
import seaborn as sbn

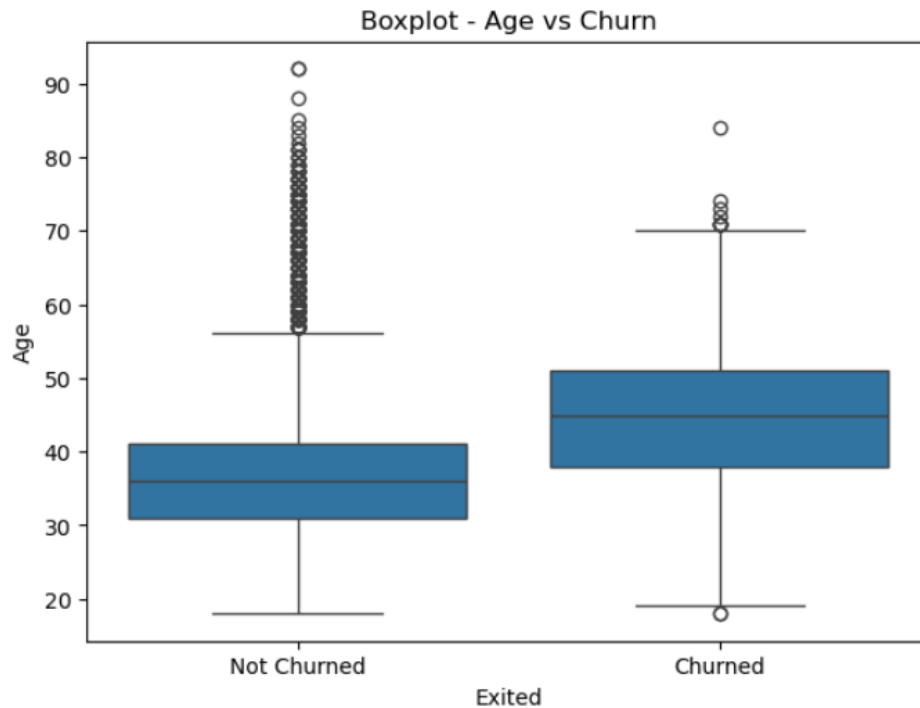
#creating group for churned and non churned
churned=df[df['Exited']==1]
not_churned=df[df['Exited']==0]

##Histogram
plt.figure(figsize=(10,5))
plt.hist(not_churned['Age'], bins=40, alpha=0.6, label='Not Churned')
plt.hist(churned['Age'], bins=20, alpha=0.8, label='Churned')
plt.title("Age Distribution by Churn Status")
plt.xlabel("Age")
plt.ylabel("Count")
plt.legend()
plt.show()
```



Most of the customers that are not churned lie within the age group of 30-45.
Churned customers lie within the age group of 40-50 .

```
[58]: # Boxplot of Age by Churn
plt.figure(figsize=(7,5))
sbn.boxplot(x='Exited',y='Age',data=df)
plt.xticks([0,1],['Not Churned', 'Churned'])
plt.title("Boxplot - Age vs Churn")
plt.show()
```



According to the boxplot outcome , the non-churned customers lie within the age group of 30-40 and the outliers exist above the age of 58 . There are the multiple outliers . And the churned customers lie within the age group of 40-55 .Whereas the outliers lie above the age of 70 .There are limited outliers that exist .

18. Is there any correlation among numeric features like CreditScore, Balance, and EstimatedSalary?

```
[64]: numeric_columns = ['CreditScore', 'Balance', 'EstimatedSalary']
correlation_matrix = df[numeric_columns].corr()
print("\nCorrelation matrix for selected numeric columns:")
print(correlation_matrix)
```

```
Correlation matrix for selected numeric columns:
          CreditScore  Balance  EstimatedSalary
CreditScore      1.000000  0.006268      -0.001384
Balance          0.006268  1.000000       0.012797
EstimatedSalary  -0.001384  0.012797       1.000000
```

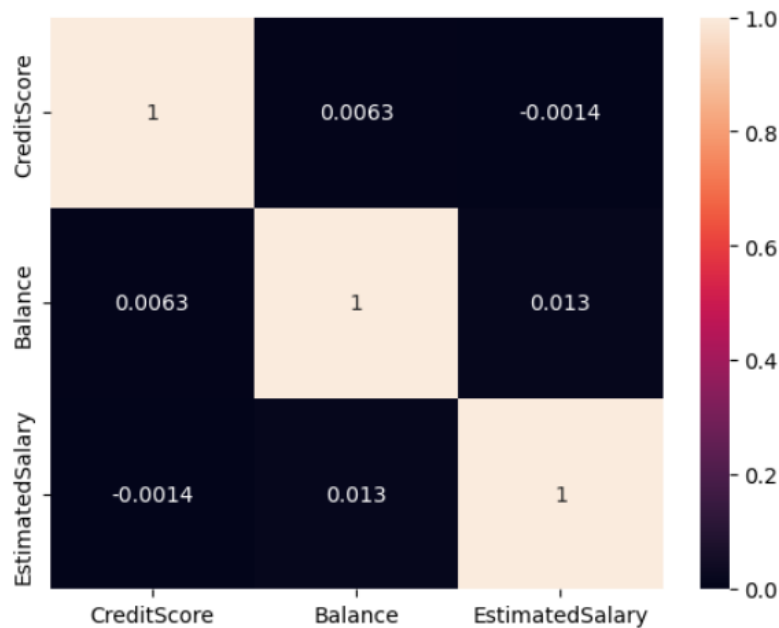
These variables do not influence each other due to no relation.

Since the correlations are nearly 0 , they do not impact the churn rate .And the variables should be examined individually to understand the churn rate rather than analyzing it via correlation .

19. What does a heatmap reveal about feature interactions with churn?

```
[65]: sns.heatmap(correlation_matrix,annot=True)
```

```
[65]: <Axes: >
```



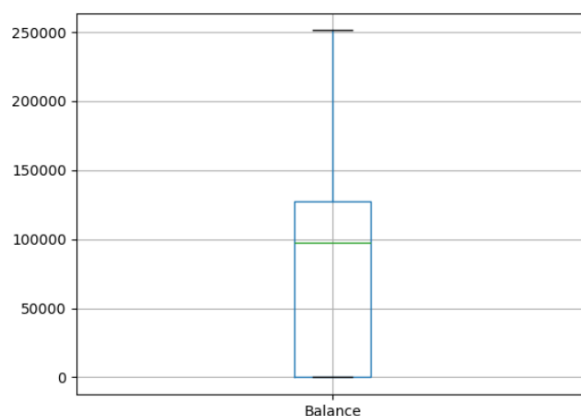
There is weak correlation across all three numeric variables .
CreditScore and Balance = 0.0063
CreditScore and Estimated Salary = -0.0014
Balance and Estimated Salary = 0.013.

20. Are there outliers in Balance, CreditScore, or Age that are mostly associated with churn?

```
[107]: #Boxplot for Balance , CreditScore, Age
```

```
[104]: df.boxplot("Balance")
```

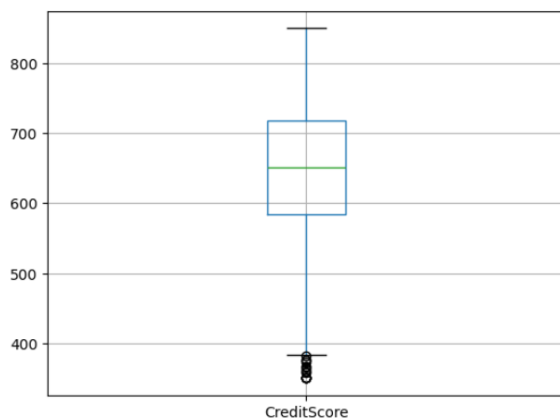
```
[104]: <Axes: >
```



There are no outliers found for Balance as seen in the figure. And many customers have 0 balance .
The bank has both 0 balance and high value customers .But there are no outliers .In real times , 0 balance customers are more likely to churn as they have no active relationship with the bank.

```
[106]: df.boxplot("CreditScore")
```

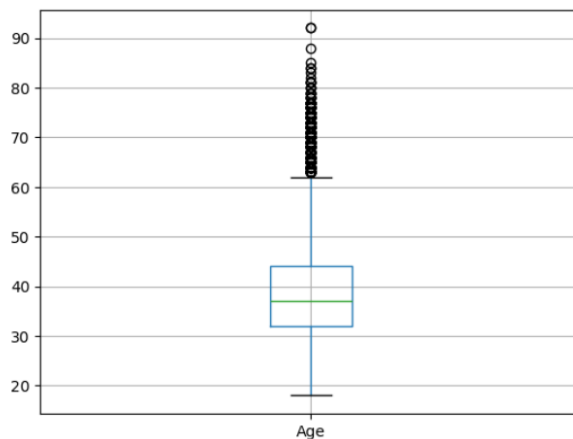
```
[106]: <Axes: >
```



Outliers exist at the lower end around 300-400. And most of the Credit Scores lie between 580-720. Very low credit scores are rare, so they appear as outliers. And in real times Customers with low credit scores outliers are more likely to churn.

```
[105]: df.boxplot("Age")
```

```
[105]: <Axes: >
```



There are many outliers above Age 60 years. And most of the customers are aged 30-45. So the Older age group people are uncommon so they are marked as outliers. Older customers have mostly higher churn rates. And they are more likely to churn.

21. Group customers into age brackets (e.g., 18-30 as Adults, 30-50 as middle age and 50-100 as seniors.). How does churn rate change across them?

```
[88]: # Creating Age Brackets to calculate churn rates for different age groups
```

```
bins=[18,30,50,100]
labels=['Adults(18-30)', 'Middle Age(30-50)', 'Seniors(50-100)']

df['AgeGroup']=pd.cut(df['Age'],bins=bins,labels=labels,right=False)
churn_rate_age=df.groupby('AgeGroup')['Exited'].mean()*100
print(churn_rate_age)
```

```
AgeGroup
Adults(18-30)      7.556368
Middle Age(30-50)  18.380241
Seniors(50-100)   45.448029
Name: Exited, dtype: float64
```

```
C:\Users\jyoti\AppData\Local\Temp\ipykernel_20268\3598292375.py:7: FutureWarning: The default of observed=False is deprecated and will be changed to True in a future version of pandas. Pass observed=False to retain current behavior or observed=True to adopt the future default and silence this warning.
  churn_rate_age=df.groupby('AgeGroup')['Exited'].mean()*100
```

Churn Rates are more i.e. 45.4 % for Senior Age group .
And least for Adults i.e. 7.5 % .
Whereas the churn rate is 18.3% for Middle Age group.

22. Are customers with only one product (NumOfProducts = 1) more likely to churn than those with multiple?

```
[109]: #Determining churned customers having only one numofprod compared to the customers having multiple .
```

```
churn_rate=df.groupby('NumOfProducts')['Exited'].mean()*100  
#churn_rate.rename(columns={'Exited':'ChurnRate'},inplace=True)  
print(churn_rate)
```

```
NumOfProducts  
1    27.714398  
2     7.603486  
3    82.706767  
4   100.000000  
Name: Exited, dtype: float64
```

Here the churn rates are calculated based on the number of products , the customers own. Customers with 1 product have high churn rates compared to those having 2 products. But here customers with 3 or 4 products show extremely high churn whereas customers with 2 products are mostly loyal and have 7.6% of churn rates .